

Lecture 4 Notes - Characteristics of Time Series

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- **Reading:** Chapter 1.3-1.7 – Shumway and Stoffer

1 Today

- Autocovariance, autocorrelation, cross-covariance, cross-correlation
- Stationarity

1.1 Covariance of linear combinations of random variables

If we have random variables $U = \sum_{j=1}^m a_j X_j$ and $V = \sum_{k=1}^r b_k Y_k$ that are linear combinations of (finite variance) random variables X_j and Y_k , then the covariance of these is:

$$\text{cov}(U, V) = \sum_{j=1}^m \sum_{k=1}^r a_j b_k \text{cov}(X_j, Y_k)$$

1.2 Autocovariance of a random walk

Recall that a random walk (with or without drift) is:

$$x_t = \delta t + \sum_{i=1}^t w_i$$

The w_t are uncorrelated random variables.

$$\begin{aligned} \gamma(s, t) &= \text{cov}(x_s, x_t) \\ &= \text{cov}\left(\delta s + \sum_{i=1}^s w_i, \delta t + \sum_{i=1}^t w_i\right) \\ &= \sigma^2 \min(s, t) \end{aligned} \tag{1}$$

In this case, the autocovariance *does* depend on the particular s and t chosen. What if we want a bounded measure?

2 Autocorrelation function

We can calculate the autocorrelation function (ACF) as:

$$\rho(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(s, s)\gamma(t, t)}}$$

This measures the linear predictability of the time series at time t (x_t) using x_s . The Cauchy-Schwarz inequality states that:

$$\text{cov}(x, y) \leq \sqrt{\text{Var}(x) \text{Var}(y)}$$

We can also extend this to looking at the linear predictability of one time series to another, by extending into the concepts of *cross-covariance* and *cross-correlation*.

3 Cross-covariance

Between two time series x_t and y_t :

$$\gamma_{xy}(s, t) = \text{cov}(x_s, y_t) = \mathbb{E}[(x_s - \mu_{xs})(y_t - \mu_{yt})]$$

This tells us how the values in y relate to the values in x over time.

Let's think about a simple example:

$$y_t = x_{t-2}$$

What is $\gamma_{xy}(k)$ (for lag k)? At what lag is γ_{xy} maximized?

We can also have the normalized version:

4 Cross-correlation

$$\rho_{xy}(s, t) = \frac{\gamma_{xy}(s, t)}{\sqrt{\gamma_x(s, s)\gamma_y(t, t)}}$$

This is bounded such that $-1 \leq \rho(s, t) \leq 1$

5 Stationarity

Recall that for a moving average, the autocovariance $\gamma_v(s, t)$ depends only on the time separation between s and t (also called the *lag*). This is important because this implies the concept of *stationarity*.

A *strictly stationary* time series is one where every collection of values has identical probabilistic behavior to the time-shifted set:

$$\{x_t, x_{t_2}, \dots, x_{t_k}\} \stackrel{d}{=} \{x_{t_1+h}, x_{t_2+h}, \dots, x_{t_k+h}\}$$

(that is, same mean, variance, higher-order moments for all t). Examples: iid process

This is not true for most applications and is too strict of a definition. So instead, we will introduce the concept of weak stationarity. In your book this is just called “stationary” as short hand.

5.1 Weakly stationary

A weakly stationary time series x_t is a finite variance process where:

- The mean function is constant and doesn't depend on t
- The autocovariance function $\gamma(s, t)$ depends on s and t only through their difference $|s - t|$.

This is convenient because we can then estimate things about time series where we don't have multiple repeated observations (and thus we can't actually estimate the variability for a given time sample directly). In a stationary time series, the mean function is independent of time, so we have:

$$\mu_t = \mu$$

We can also simplify the autocovariance function so that it is only dependent on the time shift / lag. For example, if $s = t + h$, h is the *lag* between s and t . We then have:

$$\gamma(t + h, t) = \text{cov}(x_{t+h}, x_t) = \text{cov}(x_h, x_0) = \gamma(h, 0) = \gamma(h)$$

The autocovariance of a (weakly) stationary time series is thus:

$$\gamma(h) = \text{cov}(x_{t+h}, x_t) = \mathbb{E}[(x_{t+h} - \mu)(x_t - \mu)]$$

- In your lab, you will look at the stationarity of white noise (strictly stationary), moving average (weakly stationary), random walks (not stationary), and linear trends (not stationary).
- If mean and/or autocovariance change with time, your time series is not stationary

6 Estimating covariance of a single time series

Much of the time we don't have multiple samples of our time series, so we can't estimate μ_t separately for each t . If we assume stationarity, $\mu_t = \mu$, so we can use the sample mean instead of μ_t :

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x})(x_t - \bar{x}),$$

where $\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t$ is the sample mean. Also, $\hat{\gamma}(-h) = \hat{\gamma}(h)$ for $h = 0, 1, \dots, n-1$.

This is nice because we can always calculate the sample autocovariance. However, whether it is interpretable or meaningful will depend on whether the stationarity assumption is approximately true.

7 Estimating relationships between two time series

We can use cross-correlation to estimate relationships between two series x_t and y_t . For signals that are jointly weakly stationary:

$$\hat{\rho}_{xy}(h) = \frac{\hat{\gamma}_{xy}(h)}{\sqrt{\hat{\gamma}_x(0)\hat{\gamma}_y(0)}}$$

Here is an example of the autocorrelation functions for the Southern Oscillation Index, fish recruitment, and their relationship (cross-correlation function):

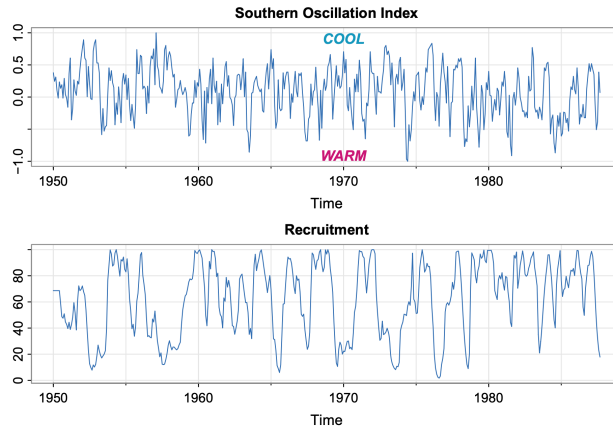


Fig. 1.5. Monthly SOI and Recruitment (estimated new fish), 1950–1987

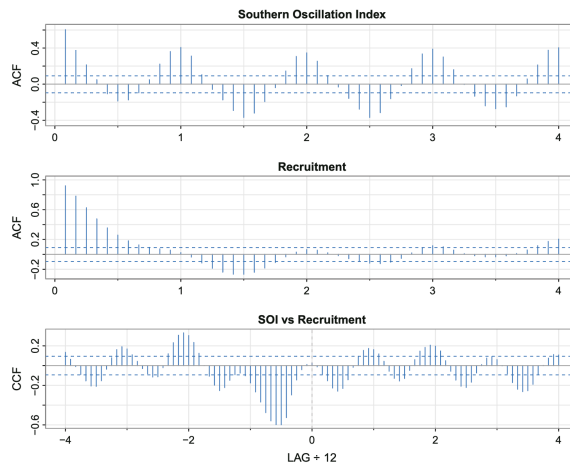


Fig. 1.18. Sample ACFs of the SOI series (top) and of the Recruitment series (middle), and the sample CCF of the two series (bottom); negative lags indicate SOI leads Recruitment. The lag axes are in terms of seasons (12 months)

8 Next time:

- Linear regression (Chapter 2 SS)